

🛡️ Capstone Presentation • DSB PT3

Detecting Phishing URLs

Real-Time Machine Learning Classification Directly from Raw URL Strings

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Introduction & Motivation

The Challenge of Reactive Defense

Traditional phishing protection relies on network lookups, WHOIS records, page rendering, or blocklists. These systems are inherently **reactive** — protecting users only after a URL has been visited and flagged elsewhere.

Zero-Network Pre-Filter

By scoring a URL using strictly its **raw text string** (length, symbols, host structure, vocabulary), detection runs instantly at the endpoint (browser, email client) before any network request or external query occurs.

Problem Framing & Core Objectives

⚠ Problem Statement

*Can we reliably predict whether a URL is a **phishing** attempt using only information in the string itself without performing DNS/WHOIS queries?*

Using a dataset of **450K** URLs with a **76.8% legitimate / 23.2% phishing** split, build a binary classifier that predicts phishing URLs from structural, lexical & NLP features — without visiting the page, targeting **90%+ accuracy** and **≥90% recall** on the phishing class, as an instant, zero-latency first-pass filter operating directly on client links.

Published benchmarks show ML-based URL detection typically achieves **90–95% accuracy**.

🎯 Core Objectives

- ✦ Engineer ≥ 24 hand-crafted features directly from raw URL strings.
- ✦ Integrate an NLP layer (character n-grams + bag-of-words) to evaluate lexical vocabulary boost.
- ✦ Train & tune Logistic Regression, Random Forest, & XGBoost using strict host-grouped validation.

Data Summary & The Data Artifact Discovery

- ◆ Dataset Size: 450,176 raw URLs (Subsampled to 150,000 for modeling tractability).
- ◆ Target Labels: 76.8% Legitimate (345,738) vs 23.2% Phishing (104,438).
- ◆ Host Repetition: 162,570 unique hosts (`en.wikipedia.org` appears 12,687 times).

Feature Engineering Architecture



Block A: Hand-Crafted

24 Features: URL/Host/Path lengths, character counts (dots, slashes, `@`, `=`), digit ratios, IP host flag, lure words, and Shannon entropy.



Block B & C: NLP Layer

TF-IDF Char N-Grams: (3-5 chars, 30,000 terms).

Bag-of-Words: Binary tokens (8,021 terms). Built strictly on de-schemed core URLs.

Exploratory Data Analysis: Key Findings



Structural Differences

Phishing URLs skew noticeably longer (mean 58.75 vs 46.49 chars), carry more subdomains (0.47 vs 0.29), and have higher hostname entropy (3.38 vs 3.31).



Obfuscation Signals

Phishing URLs show higher digit counts (7.42 vs 3.28), frequent IP host usage (4.1% vs 0%), explicit port numbers, and heavy use of URL shorteners.



NLP Token Dynamics

Chi-square analysis highlights strong phishing signals in WordPress compromise vocabulary: wp, includes, plugins, admin, and login.

Model 1: Logistic Regression (Best Model)

0.891

Test F1 Score

0.985

ROC-AUC

0.965

PR-AUC

$$\text{logit}(p) = \beta_0 + \sum_{i=1}^{38,045} \beta_i x_i \quad \Rightarrow \quad p = \frac{1}{1 + e^{-\text{logit}(p)}}$$

Hyperparameter Search: RandomizedSearchCV with 3-Fold Host GroupKFold.

Best Parameters: C = 10, penalty = 'l2', solver = 'liblinear', class_weight = 'balanced'.

Threshold Tuning: Optimal F1 threshold found at $p \geq 0.596$.

Performance: Precision = 0.859, Recall = 0.924, Accuracy = 0.953.

Model 2: Random Forest Classifier

0.844

Test F1 Score

0.884

Highest Precision

0.961

ROC-AUC

⚙️ Best Parameters & Setup

```
n_estimators = 150 max_depth = None (Full depth)
min_samples_leaf = 1 max_features = 'sqrt' class_weight =
'balanced' feature_set = Reduced (424 columns)
```

📈 Performance Metrics

- ♦ F1-Optimal Threshold: $p \geq 0.540$
- ♦ Precision: 0.884 (Best precision across 3 models).
- ♦ Recall: 0.807 | Accuracy: 0.938
- ♦ PR-AUC: 0.905

Model 3: XGBoost Gradient Boosting

0.861

Test F1 Score

0.869

Balanced Recall

0.972

ROC-AUC

Best Parameters & Setup

```
max_depth = 10 learning_rate = 0.2 n_estimators =  
600 subsample = 0.9, colsample_bytree = 1.0  
scale_pos_weight = 3.5 (Train ratio) tree_method = 'hist'
```

Performance Metrics

- ✦ F1-Optimal Threshold: $p \geq 0.665$
- ✦ Precision: 0.853 | Recall: 0.869
- ✦ Accuracy: 0.942
- ✦ PR-AUC: 0.934

Model Comparison & Performance Matrix

Model	Threshold	F1 Score	Precision	Recall	Accuracy	ROC-AUC	PR-AUC
Logistic Regression (Best)	0.596	0.891	0.859	0.924	0.953	0.985	0.965
Random Forest	0.540	0.844	0.884	0.807	0.938	0.961	0.905
XGBoost	0.665	0.861	0.853	0.869	0.942	0.972	0.934
Naive Baseline (Always Legit)	0.500	0.000	0.000	0.000	0.793	0.500	0.220

Evaluation Context: Evaluated on 30,924 untouched test URLs using GroupShuffleSplit (0% host overlap). Thresholds tuned on training probabilities.

Why Logistic Regression Won & Key Takeaways

🏆 Why the Linear Model Won

- ◆ High-Dimensional Advantage: Operates natively on the full 38,045 sparse feature space without feature truncation.
- ◆ Tree Constraint: Tree models required Chi-Square truncation to 400 NLP columns for compute tractability, discarding nuanced vocabulary interactions.
- ◆ Linear Separability: Character n-grams plus structural indicators form a strongly linearly separable decision boundary.

🚀 Business & Deployment Impact

- ◆ Sub-Millisecond Inference: Scoring requires only vector dot-products — ideal for browser extensions & endpoint security.
- ◆ High Recall Filter: Catches 92.4% of phishing attempts at first-pass before network calls occur.
- ◆ Interpretable Coefficients: Fully audit-ready feature weights for security analysts.

 Live Demo: Phishing URL Inspector

Questions & Discussion

A fast, zero-lookup URL classifier serving real-time predictions via Flask & Joblib bundle.

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